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Cloud detection algorithm for multi-modal satellite imagery using convolutional neural-networks (CNN)

Michal Segal-Rozenhaimer[a,b,c,*], Alan Li[a,b], Kamalika Das[d,1], Ved Chirayath[a,**]

[a] NASA Ames Research Center – Moffett Field, CA, USA
[b] Bay Area Environmental Research Institute – Moffett Field, CA, USA
[c] Department of Geophysics, Porter School of the Environment and Earth Sciences, Tel-Aviv University, Tel-Aviv, Israel
[d] VMWare Inc., CA, USA

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ABSTRACT

Cloud detection algorithms are crucial in many remote-sensing applications to allow an optimized processing of the acquired data, without the interference of the cloud fields above the surfaces of interest (e.g., land, coral reefs, etc.). While this is a well-established area of research, replete with a number of cloud detection methodologies, many issues persist for detecting clouds over areas of high albedo surfaces (snow and sand), detecting cloud shadows, and transferring a given algorithm between observational platforms. Particularly for the latter, algorithms are often platform-specific with corresponding rule-based tests and thresholds inherent to instruments and applied corrections. Here, we present a convolutional neural network (CNN) algorithm for the detection of cloud and cloud shadow fields in multi-channel satellite imagery from World-View-2 (WV-2) and Sentinel-2 (S-2), using their Red, Green, Blue, and Near-Infrared (RGB, NIR) channels. This algorithm is developed within the NASA NeMO-Net project, a multi-modal CNN for global coral reef classification which utilizes imagery from multiple remote sensing aircraft and satellites with heterogeneous spatial resolution and spectral coverage. Our cloud detection algorithm is novel in that it attempts to learn deep invariant features for cloud detection utilizing both the spectral and the spatial information inherent in satellite imagery. The first part of our work presents the CNN cloud and cloud shadow algorithm development (trained using WV-2 data) and its application to WV-2 (with a cloud detection accuracy of 89%) and to S-2 imagery (referred to as augmented CNN). The second part presents a new domain adaptation CNN-based approach (domain adversarial NN) that allows for better adaptation between the two satellite platforms during the prediction step, without the need to train for each platform separately. Our augmented CNN algorithm results in better cloud prediction rates as compared to the original S-2 cloud mask (81% versus 48%), but still, clear pixels prediction rate is lower than S-2 (81% versus 91%). Nevertheless, the application of the domain adaptation approach shows promise in better transferring the knowledge gained from one trained domain (WV-2) to another (S-2), increasing the prediction accuracy of both clear and cloudy pixels when compared to a network trained only by WV-2. As such, domain adaptation may offer a novel means of additional augmentation for our CNN-based cloud detection algorithm, increasing robustness towards predictions from multiple remote sensing platforms. The approach presented here may be further developed and optimized for global and multi-modal (multi-channel and multi-platform) satellite cloud detection capability by utilizing a more global dataset.

1. Introduction

Cloud detection is one of the most critical steps in pre-processing of optical satellite images. Unsuccessful cloud and cloud shadow masking, which obscures, brightens or darkens specific scenes, respectively (Zhu and Woodcock, 2012), will cause subsequent negative effects on many of the traditional remote sensing tasks such as change and anomaly detection, segmentation, and classification (e.g., land-use or marine habitats as in the work presented here), crop monitoring tasks and others. However, detection of clouds and cloud shadows is still not

* Corresponding author. NASA Ames Research Center – Moffett Field, CA, USA.
** Corresponding author.
E-mail addresses: michal.segalrozenhaimer@nasa.gov (M. Segal-Rozenhaimer), ved.c@nasa.gov (V. Chirayath).
[1] Work was done while the author was at USRA (Universities Space Research Association), at the NASA Ames Research Center.

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straightforward, owing to their high spectral heterogeneity and the spectral and temperature variability of the underlying surface (Zhu and Woodcock, 2012). The main aspect of this difficulty is attributed to the fact that most of the existing operational algorithms are rule-based, which means that their tests of detecting clouds depend on specific intensity thresholds and rules that are carefully tailored for a given satellite platform and sensors. Such an approach is also sensitive to variations in atmospheric conditions and to absolute surface reflectance values. The latter is also critical for the success rate for detecting cloud shadows, which can also resemble the spectral signature of terrain shadows, dark vegetation and water bodies, where if depends on absolute values can confuse the algorithm (Hughes and Hayes, 2014). The absence of thermal infrared bands in the older Landsat versions (1–5), Sentinel-2 and WorldView-2 is another reason why such satellites are performing suboptimal in comparison with moderate-spatial-resolution imagers such as MODIS (Moderate Resolution Imaging Spectroradiometer). As Landsat-7 has one relatively broad thermal IR band and Landsat-8 has a SWIR cirrus band and two thermal-IR bands with the TIRS (Thermal Infrared sensor), these are now being utilized in its cloud detection schemes mentioned below, showing improvement over former capabilities. Among the remote sensing satellite imagers, Landsat has the longest heritage of automated detection algorithms of clouds and cloud shadows. Currently, available algorithms for Landsat include Landsat 7 Automated Cloud Cover Assessment (ACCA) (Irish, 2000; Irish et al., 2006), Landscape Fire and Resource Management Planning Tools (LANDFIRE) CCA (Reeves et al., 2006), Landsat Ecosystem Disturbance Adaptive Processing System (LEDAPS) cloud algorithm (Vermote and Saleous, 2007), Function of Mask (Fmask) algorithm (Zhu and Woodcock, 2012), and Tmask (multiTemporal mask) (Zue and Woodcock, 2014), which use a mixture of threshold based tests, and include both visible and thermal IR bands. All the aforementioned algorithms are single-scene algorithms, except for Tmask, which uses time-series imagery for the differentiation of clouds and land surfaces. The latter method is in general superior to single-scene methods; however, it requires multiple images of the same location, and a reference cloud-free image, which is often hard to obtain. ACCA serves as the operational Landsat TM and ETM + cloud detection algorithm (Arvidson et al., 2006), with an accuracy of approximately 83% (Foga et al., 2017). Of the aforementioned algorithms, Fmask possesses the highest cloud mask detection accuracy of approximately 90% (Foga et al., 2017), and was also modified to work with Sentinel-2 images (Zhu and Woodcock, 2012), but does not perform as well due to the lack of thermal-IR bands in Sentinel-2. In addition, there are algorithms that were specifically developed for Sentinel-2 and include the European Space Agency (ESA) Sen2Cor (http://step.esa.int/main/third-party-plugins-2/sen2cor/) software package, which performs atmospheric correction together with cloud masking using auxiliary data information from land-use and climate change databases. Although this single-scene algorithm is widely used, it still encounters issues of over/underestimation of cloud cover over water. Another available algorithm for Sentinel-2 is MAJA (Maccs-Atcor Joint Algorithm), which is based on time-series analysis (i.e. multi-scene algorithms), and is developed, processed and maintained by the European space agency (ESA) community (Lonjou et al., 2016). However, it is only processed for specific areas so far since it requires a large amount of imagery and computational resources. Indeed, machine-learning-based approaches necessitate large computational resources as well, although this is an issue predominantly during the training stage. Overall, the majority of the operational cloud detection algorithms to date involve the application of static and dynamic rule-based tests for their operation, which makes the transfer of the algorithm from one sensor to another a tedious and, at times, impossible task. As machine learning is gaining popularity, several non-operational algorithms exist that utilizes various methods to detect clouds from Satellite imagery. Hughes and Hayes (2014), designed a neural-network (NN) predictor that ingests band information together with pre-calculated spatial information from the imagery to detect clouds and their shadows. However, their features rely on prior knowledge and pre-processing of the image before it gets introduced into the network for prediction. Another machine learning approach for cloud detection only was used by Wang et al. (2008) using support vector machines (SVM) to MODIS multi-band (including the thermal infrared bands) imagery. This approach has shown relative success, however, as this is a linear classifier, it has limited applicability. More recent work of utilizing convolutional NN (CNN) is documented in Zi et al. (2018), Chen et al. (2018) and in Weiland et al. (2019). CNNs are a newer form of NN, where their architecture allows the direct ingestion of 3D inputs (e.g. Maggiori et al., 2017) such as images and the operations within the network architecture are made over the spatial and spectral domains concurrently through volume (3D) operations rather than on a single neuron like in traditional NNs schemes. The convolution operations preserve the spatial relationship between pixels by learning image features using collocated grids of input data. On each layer many different types of convolution operations can be applied, which mimic a set of various filters (extracting various features) applied onto the data volume. This approach supersedes many other methods by eliminating the need to selectively predesign specific filters (or thresholds) specific for the task at hand, while the network is trained to learn the best filter and threshold values needed for the selected task.

Zi et al. (2018) and Chen et al. (2018) used super-pixel segmentation prior to their introduction into a CNN scheme for cloud detection of Landsat and the high-resolution Chinese ZY-3, Gaofeng-1 and Gaofeng-2 satellites. Even though the segmentation methods might result in slightly different end-results, the latter work have shown great promise and high accuracy rates compared to the traditional NN, SVM and K-means methods. Another recent work (Weiland et al., 2019) utilizing CNNs to detect cloud and cloud shadow for the Landsat and Sentinel sensors have shown that such algorithms are comparable and can be even superior to the traditional algorithms such as Fmask. In this work, the algorithm has utilized the RGB, NIR and SWIR channels, with the latter missing from the WV-2 satellite. Hence, the current work attempts to optimize this approach for our purposes by reducing the amount of channels utilized for the cloud and cloud shadow classification task. Also, although Weiland et al. (2019) discussed multi-sensor generalization ability, they only tested their algorithm among various Landsat sensors and the Sentinel sensor, which have very similar band locations and widths, unlike the work shown here transferring the algorithm between Sentinel and WV-2.

Our work of developing a new cloud and cloud shadow products for the high-resolution WV-2 satellite stemmed from the need to support the NASA's NeMO-Net, The **Ne**ural **M**ulti-Modal **O**bservation & Training **Net**work for Global Coral Reef Assessment project work (http://nemonet.info). This project aims at assessing the present and past dynamics of shallow marine ecosystems, particularly coral reefs, through determination of percent living cover, mapping of spatial distribution, and segmentation of geomorphology and biology from fusion of remote sensing data from multiple airborne and satellite platforms (Chirayath et al., 2018a, 2018b; Chirayath, 2016; Chirayath and Earle, 2016; Chirayath and Instrella, 2018; Chirayath and Li, 2018). The fusion of high-resolution imagery from WV-2 and airborne instruments is used in the learning process and the results are applied onto global, but lower resolution platforms, such as Sentinel-2 and Landsat. As clouds and their shadows interfere with our marine habitat classification work, and such products are not available for the WV-2 sensor, we embarked on developing this new product. In addition, the project's need to fuse WV-2 with Sentinel-2 and later-on Landsat imagery for assessing global marine habitats health motivated our attempt to develop a product that can be easily applied onto multiple platforms without the need for its specific tailoring to each platform.

Since CNNs in essence are able to extract both the spectral and spatial features from the raw multi-channel imagery, they are capable of identifying the cloud and cloud shadow classes even when not all the

relevant channels are available (i.e. SWIR and thermal channels that are missing from the very high-resolution satellites such as WV-2 and others). Hence, for the purpose of our study, which uses WV-2 imagery, the approach of utilizing the CNN architecture seemed to be the most appropriate.

We present here a machine learning method using CNNs for cloud and cloud shadow detection, developed specifically for the high-resolution (~2 m) WV-2 imagery using only RGB and NIR bands, as these channels are also shared among Sentinel-2 and Landsat.

We further explore and expand upon "cross-platform" applicability potential by using a combination of CNN and domain adversarial NN (DANN) methods to allow for better transferability of our cloud detector mask from WV-2 to Sentinel-2. Such an approach can eliminate the need of a special adaptation procedure between different platforms, as was done by the Fmask algorithm transferred from Landsat to Sentinel-2 (Zhu and Woodcock, 2012).

This manuscript is organized in the following manner: Section 2 outlines the study area, focusing on the South Pacific Ocean, and the WV-2 and S-2 datasets, section 3 describes the methodology and the training and test data used for our algorithm. Section 4 provides the results of the cloud mask only and the cloud mask combined with cloud shadow detection algorithms applied on WV-2 imagery, and also the cloud masking as applied to Sentinel-2 imagery. We also present results of the prediction of cloud masking on Sentinel-2 data after applying a DANN approach. Section 5 discusses the strengths and weaknesses of the new method, and section 6 concludes with future development that can be implemented by the community so that this approach can be applied to a global dataset.

1.1. Data

1.1.1. Study area

Our main study area is situated in the Fiji Islands in the Southern Pacific Ocean. This was used as the primary site in which we are applying our NeMO-Net marine habitat and coral reef classification system. The area encompasses longitudes between 177 and 179°W and latitudes between 16 and 19°S, as shown in Fig. 1 (top panel, see island locations marked on map). Specific islands that were used for the analysis of both the marine habitat classification and the cloud algorithm are highlighted in Fig. 1 (bottom panel).

WorldView-2 (WV-2) is one of the satellites within DigitalGlobe (www.digitalglobe.com). Launched in October 2009, is the first high-resolution 8-band multispectral commercial satellite. Operating at an altitude of 770 km, at a sun-synchronous orbit, WorldView-2 provides 46 cm panchromatic resolution and 1.85 m multispectral resolution, which is unprecedented, and thus is especially well suited for marine and coral habitat classification tasks. WorldView-2 has an average revisit time of 1.1 days and is capable of collecting up to 1 million km^2 of 8-band imagery per day. While only territorial US imagery is available on-line free of charge from the USGS data archives, and some additional data is available through the European Space Agency (ESA), for specific projects, the majority of the data is for commercial use and must be purchased.

For our NeMO-Net (Chirayath et al., 2018a) project, we leveraged data from the Living Oceans Foundation's Global Reef Expedition (Purkis et al., 2018) to implement our algorithms. The foundation led a global reef expedition (https://www.livingoceansfoundation.org/global-reef-expedition/) between 2011 and 2016, where reef images and ground-truthing were taken from a ship vessel around the oceans of the world, including the Atlantic/Caribbean Ocean in 2011–2012, Pacific Ocean (including the South Pacific) in 2012–2015 and the Indian Ocean in 2015. To complement the ground-truthing and in-situ sampling efforts, satellite imagery from WV-2 were acquired over the expedition regions. The expedition provided high-resolution habitat and bathymetry maps for over 65,000 square kilometers of Earth's remotest coral reefs. These data are highly unique and valuable as there are very few datasets with such extensive global coverage and ground-truthing for 24 marine habitat and coral reef geomorphological classes. As part of the NeMO-Net project, we have evaluated the dataset from the Indian Ocean (Peros Banhos) and Fiji Islands in the South Pacific Ocean. As the latter dataset is much more comprehensive (covers multiple islands and spans different satellite acquisition times between 2010 and 2013) we have focused on this dataset in our NeMO-Net algorithm development (Chirayath et al., 2018b), and also in our cloud detection algorithm development presented here.

While this dataset includes images classified using an object segmentation software, verified by multiple ground-truth diver classification and manually inspected by marine habitat experts, it does not include a cloud or cloud shadow classes. Since both clouds and cloud shadows interfere with our classification of the marine habitat classes, and since WV-2 does not have any established cloud mask or cloud shadow mask algorithms, the algorithm presented here was specifically developed to be ingested in our pre-processing scheme for the NeMO-Net classification project.

The imagery used for the development of the cloud detection algorithm was similar to the imagery used to develop the marine habitat classifier network. In short, we have 8-channel (covering spectral range between 400 and 1040 nm) imagery for 11 of the Fiji Islands (as shown in Fig. 1 bottom panel). For each of the islands there are multiple (between 3 and 7) different acquisition times between the years of 2010 through 2013. Each overpass included between 6 and 9 tiles that composed one of each of the islands (hereafter an island mosaic). The data was converted from the raw radiometrically corrected (gain and offset corrected and dark subtracted) detector counts, $q_{pixel,Band}$ to top-of-atmosphere spectral radiance, $L_{pixel,Band}$ [in $Wm^{-2}sr^{-1}\mu m^{-1}$] according to:

$$L_{pixel,Band} = q_{pixel,Band} * K_{Band}/\Delta\lambda_{Band} \quad (1)$$

where K_{Band} is the absolute radiometric calibration factor [$Wm^{-2}sr^{-1}count^{-1}$], and $\Delta\lambda_{Band}$ is the effective bandwidth [μm] for a given band. The absolute radiometric calibration factor is unique for each set of acquisition dates and has to be applied to compare acquisitions made at different dates. In addition, to account for the different solar geometry for the different acquisition dates and times another correction was applied as follows:

$$L'_{pixel,Band} = L_{pixel,Band} * dES^2/\mu \quad (2)$$

where $L'_{pixel,Band}$ is the solar geometry corrected top-of-atmosphere band-average spectral radiance pixel, d_{ES} is the sun-earth distance at a given date and μ is the cosine of the solar zenith angle at the time of acquisition. A sample solar geometry corrected top-of-atmosphere RGB mosaic of Cicia island from WV-2 is shown in Fig. 2 left panel.

Sentinel-2 is a wide-swath, high-resolution, multi-spectral imaging mission, supporting Copernicus Land Monitoring studies, including the monitoring of vegetation, soil and water cover, as well as observation of inland waterways and coastal areas. The Sentinel-2 Multispectral Instrument (MSI) samples 13 spectral bands (covering the spectral range between 443 and 2190 nm): four bands at 10 m, six bands at 20 m and three bands at 60 m spatial resolution, from which we are utilizing initial high-resolution bands (RGB and NIR).

Corresponding Sentinel-2 images for the Fiji Islands were freely downloaded from the Sentinel-2 hub website located on Amazon AWS (https://registry.opendata.aws/sentinel-2/). To be compatible with the WV-2 data files, we downloaded Level-1C products (4-channel, including the RGB and NIR bands, as detailed in Table 1), which provide orthorectified Top-Of-Atmosphere (TOA) reflectance, with sub-pixel multispectral registration. Cloud and land/water masks are included in this product as well. The processing of Sentinel-2 Level-1C products is based on the Level-1B product (radiometrically corrected radiance data) and includes radiometric and geometric corrections including orthorectification and spatial registration on a global reference system

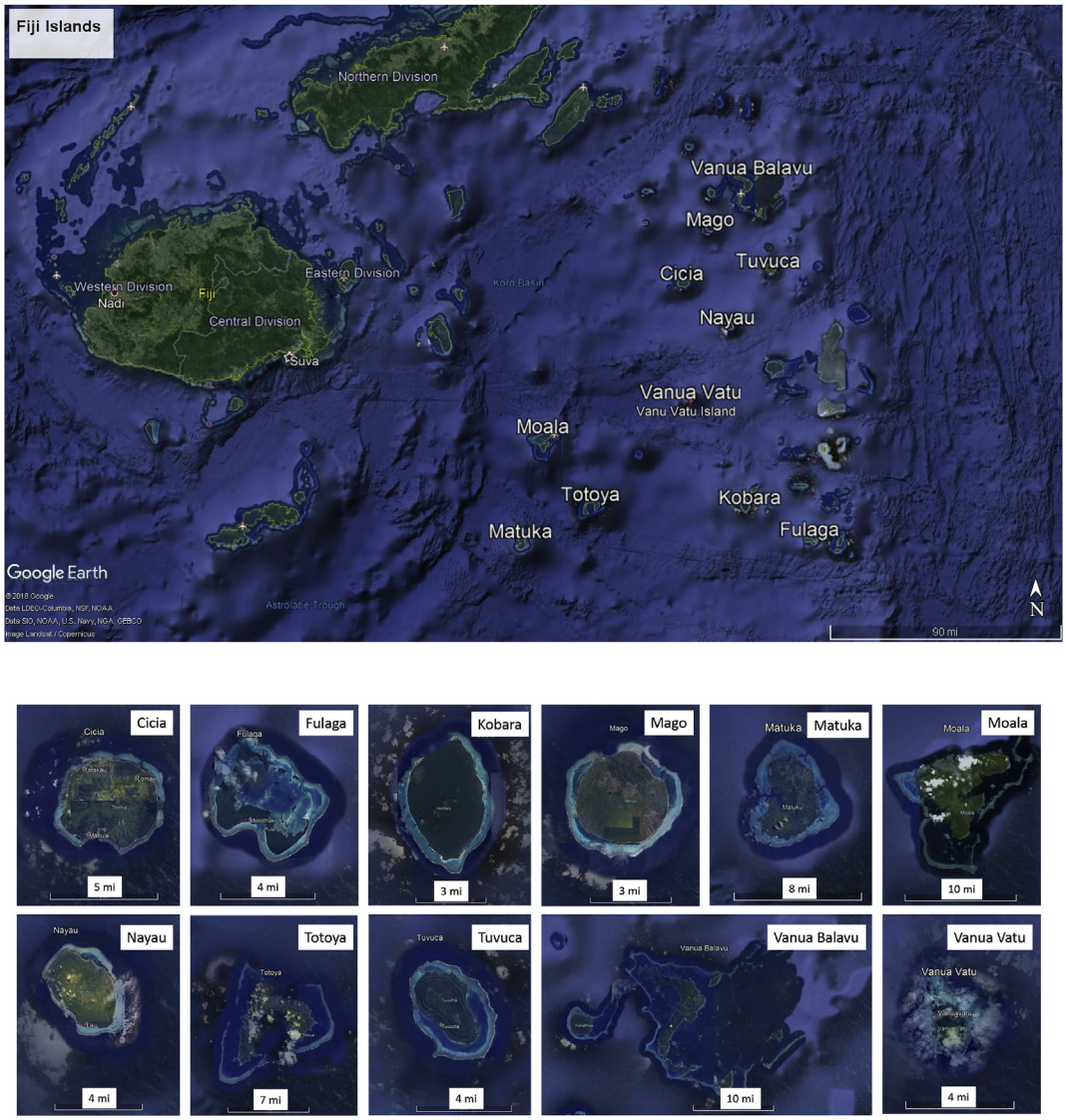


Fig. 1. (top panel) Study area overview: Fiji Islands over the South Pacific Ocean, with the islands used in the development of the cloud algorithm in bold white letters, and (bottom panel) the individual islands (zoomed-in, scale~4–10 miles).

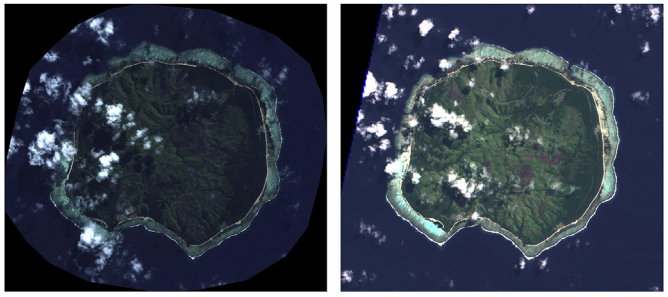


Fig. 2. (left panel) RGB solar geometry corrected top-of-atmosphere radiances mosaic of Cicial Island from WV-2, acquired on 2010-08-17, and (right panel) RGB image of Cicia Island from Sentinel-2 acquired on 2016-09-13.

with sub-pixel accuracy. Fig. 2 (right panel) shows RGB Sentinel-2 image example from 2016 to 09-13 Cicia island. Note the different spatial resolution between WV-2 and Sentinel-2 (~2 m versus 10 m, respectively), as depicted in Table 1.

Table 1
WV-2 and Sentinel-2 bands used in our cloud mask algorithm.

Satellite Name	WorldView-2	Sentinel-2
Band Name	(Band #) - Spectral range (nm) – [Spatial resolution (m)]	
Red	(5) – 630–690 – [2]	(4) – 650–680 – [10]
Green	(3) – 510–580 – [2]	(3) – 542–578 – [10]
Blue	(2) – 450–510 [2]	(2) – 475–523 – [10]
NIR	(7) – 770–895 – [2]	(8) – 784–900 [10]

2. Methods

2.1. Convolutional Neural Networks (CNN)

Neural Networks (NN) are a form of machine learning, which uses connecting computational nodes called neurons to describe in essence any linear or non-linear function. They are built upon a scheme mimicking our neural brain structure, which allows to infer/classify an output based on given inputs, but without using an implicit formula (Blackwell, 2005). A common structure for a NN is shown in Fig. 3, where the nodes act as computational elements, and are organized in hierarchical order, such that each layer of nodes (which represent a

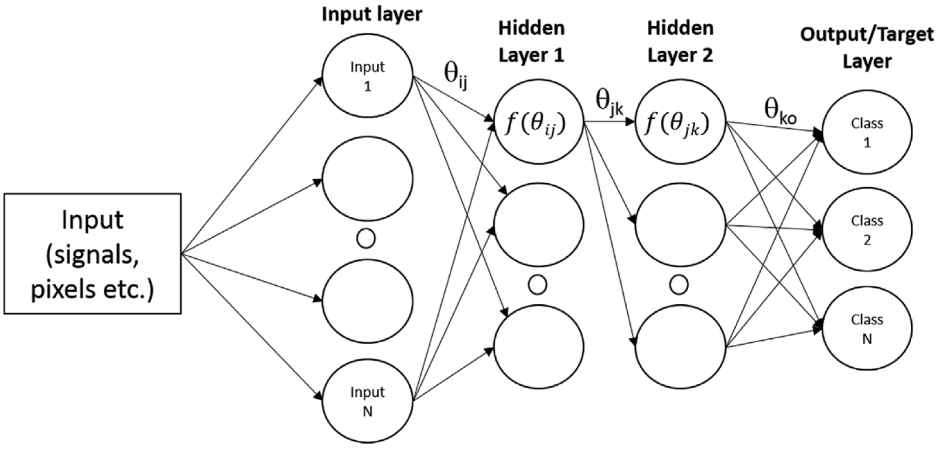


Fig. 3. Multilayer perceptron (MLP) Neural-Network (NN) standard scheme architecture, composed of only fully connected (FC) layers, with a vector of input layer, two hidden layers and one multi-class output layer. θs represent the weights that need to be determined through the training process, and *f*(θs) represent the activation function in each of the nodes.

fully connected – FC layer) allows for an increased computational complexity that can represent the system's function. The set of layers, which include the input, hidden layers (can be more than one) and the output layer represent a commonly used FC MLP (Multi-Layer Perceptron) NN architecture (LeCun, 1988; 1989, 1998). The traditional NN scheme ingests as inputs a vector of features, which, for the case of image processing purposes, can be extracted from multi-channel imagery (e.g. single pixels' intensities or radiances, derived quantities as feature vectors such as NDVI values, or principal components that are extracted from the data). The neurons representing this input vector are then each individually connected to each of the neurons in the next layer and so on up to the output layer. This means that in order to process a large input vector (which might represent a full image or a large image patch), many parameters (i.e. the connecting weights between the neurons) have to be determined during the training process. For example, for an image patch of 32 × 32 pixels with only three channels, the number of weights connecting the input and one hidden neuron to be determined through the training and optimization process would be 3072. In our case, using image patches of 256 × 256 with four channels, this amounts to 262144 weights, for only one connecting neuron (not considering biases). With many more connecting neurons in the hidden layers, this scales up quickly, which can lead to high redundancy in the network, to overfitting, and also to difficulties in convergence of the network through the optimization process (Del Frate et al., 2005; Di Noia et al., 2015). In contrast, CNN (Convolutional Neural Network) architectures (e.g., LeCun et al., 2015) are better suited to ingest and process images, as their input and hidden layers structure consists of neuron layers that are arranged in three dimensions: width, height and depth, to better fit the processing of large multi-channel, large image sets (see example schemes in Fig. 4). The neurons in a layer are then connected only to a small region of the layer before it goes through the convolution process (spatial operation on small parts of the input image), which tremendously reduces the amount of weights to be determined and their redundancy.

Another aspect differing between traditional NNs and CNNs is the ability of the latter, through their convolutional filters with trainable weights, to incorporate the feature extraction process in the training, determining the feature extraction filter that is most suitable for the task at hand. This capability is useful for two reasons. First, the convolution preserves the spatial relationship between the pixels it operates on, thus learning image features using 3D blocks of input data. This reduces the need to develop specific spatial-context based features as a pre-processing step, as was done, for example, by Hughes and Hayes (2014) in their application of traditional NN to infer cloud and cloud shadows from LandSat imagery. Second, each convolution (which can contain multiple type of convolution operations determined by the number of filters) layer extracts features of increasing complexity, effectively acting like filter banks (e.g. mean, median, maximum, low-pass, Gabor etc.) used in heritage image processing approaches (Jensen, 2016; Fogel and Sagi, 1989). In this new approach many such filters can be applied concurrently during the optimization process.

In addition, the operations within the network architecture are made along both the spatial and spectral domains through the convolution filters and pooling operations. A convolution preserves the spatial relationship between pixels by learning image features (e.g. smoothing, edge detection etc.) using blocks of input data. A pooling layer (also called subsampling or downsampling) reduces the dimensionality of each feature map but retains the most important information, by applying a variety (e.g. max, avg. or sum) of data aggregation operations. The number of different convolution and pooling operations, as shown in Fig. 4 are represented by the number of "filters" applied in each layer, similar to traditional image processing filter banks, which are manually-designed. The difference between the traditional filter banks and the convolution and pooling filters in the CNN approach is that the type of convolution and pooling operations are being learned during the training process.

For classification purposes, there are in general two main CNN approaches: (1) Image Classification, which maps each pixel (or image patch) into a label. This can be a single or multi-label classification scheme, as shown in Fig. 4 (upper panel).Label classification for the entire image patch is common in many traditional CNN applications (e.g. Vallet and Sakamoto, 2015) and the classification of the center pixel of the input image patch is the approach relevant in our case (i.e. classifies whether the center pixel of a patch shown to the network is cloudy or cloud-free for example), and (2) Semantic Segmentation, which directly returns an image that is already grouped into cloudy regions, clear regions and shadowed areas (in our case) while also accounting for the neighboring semantic pixel classes. In Fig. 4 upper panel, which represents the first CNN approach, the input image is downsized, going through the convolution, pooling and fully connected layers to the output of a class (or multi-class). In this architecture, as the layers deepen (more layers), the output is downsized compared to the input image. In approach 2 (Fig. 4 lower panel), the output size is similar to the input size by width and height and has the depth dimension of the number of classes, i.e. each class is encoded as a vector that has a value of 1 in a certain position and 0 in all other positions. In our work below, we have explored both approaches for the cloud and shadow CNNs and the DANN networks.

2.1.1. Cloud mask and cloud shadow CNN architectures

In our cloud masking work, we have explored three network architectures, following the description above: (1) MLP-NN scheme, (2) center-pixel CNN classification scheme, and (3) semantically-segmented scheme. The input into our networks is the multi-channel (RGB + NIR) image patches from the WV-2 Fiji dataset imagery. The MLP-NN architecture is similar to the one shown in Fig. 3 but has four

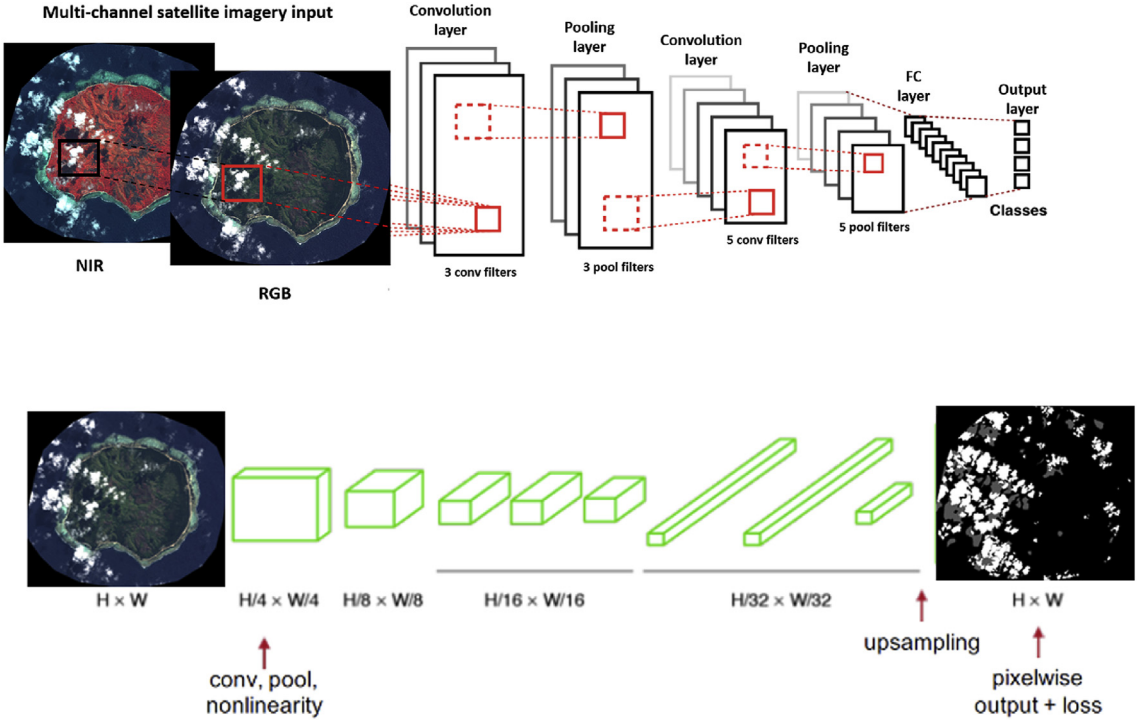


Fig. 4. (upper panel) a generic CNN pixel-wide classification architecture, which ingests both the pixel and spatial information from a multi-channel imagery (RGB + NIR) through a series of 2D convolutional layers, where the number of different convolution operations are represented by the number of filters in each layer, followed by pooling layers (mean/max operators for spatial dimensionality reduction), which can be repeated as often as needed to form a shallower or deeper architecture, a fully connected (FC) layer(s); and an output (multi-class classification in our case) layer, which can be a logistic regression (for binary classification), SOFTMAX (multi-class) or other classification output probability function. This type of architecture results in the classification of the center pixel of each input patch, and (lower panel), a semantic segmentation network, which inputs a multi-channel imagery (shown is RGB image), followed by 8 convolution (+pooling) layers, which reduces the input image size due to the pooling layers (only H and W, not D) as denoted. In this architecture the last three layers are 1 × 1 convolutions and not fully connected layers, hence, the output here is not a single label, although it is not in dimensionally similar to the input image. To achieve similar dimensionality, an upsampling operation is added to achieve the final semantic segmented output layer (cloud-free, cloud and cloud shadow in our case).

hidden layers, with 64, 64, 128, 128 nodes in each, respectively and a final layer of 2 that represents the two classes (cloud-free/cloud), using a softmax output layer, with cross-entropy as our loss function, as detailed below. This architecture was chosen to optimize prediction results versus processing time and convergence. The center-pixel CNN classification scheme followed the VGG-16 architecture by Simonyan and Zisserman (2015), with five convolutional blocks, containing increasing amount of filters in each (64, 128, 256, 512, 512), followed by normalization and pooling layers in each of the blocks and two fully connected layers, with 4096 nodes in each, for the two class centeral-pixel classification. In more detail, there are five convolutional "mini-blocks" that incorporate two consecutive convolution and activation layers, followed by a batch normalization (BN) layer and a max pooling layer. In the activation layer, we make use of the Rectified Linear Unit (ReLU) function, which takes the max between zero and the pixel value. This activation function is widely used due to its simplicity and its ability to greatly accelerate the convergence of stochastic gradient descent algorithms and their variations in training deep CNNs (Krizhevsky et al., 2012). The purpose of the BN layer is to increase the stability of a neural network, by subtracting the batch mean and dividing by the batch standard deviation. By applying this transformation, it helps to keep the inputs into the activation layer stable (not too high and not too low), and eventually maintains the mean activation close to 0 and the activation standard deviation close to 1, to help in the network training convergence (Ioffe and Szegedy, 2015). Finally, the semantic segmention architecture follows the DeepLab network architecture (Chen et al., 2017), and assigns a class to each of the pixels in the input image patch, considering the pixel neighborhood context. The DeepLab network scheme is depicted in Fig. 5. In this architecture Block#1 is similar to the VGG-16 architecture (Simonyan and Zisserman, 2015) with five mini-convolutional blocks followed by Block#2, which performs parallel operations of sparse convolutions using different dilation rates, with a fixed convolution size of 3 × 3 (following Chen et al., 2017). This second block capitalizes on both the near-and-far spatial context fields (using dilated filters of varying rates). The results from these different spatial fields (i.e. FOV's) are summed and up-sampled to expand back the redacted patch size (due to the convolution and pooling operations) via upsampling in Block#3. This approach results in an output dimension array that is similar to the 2D input size (in width and height), where the depth dimension (initially the channel dimension) has the size of the number of classes. Hence, each pixel now has a probability for each of the classes, where the highest probability dictates the assigned label rather than one label is assigned to the entire patch.

2.1.2. Training and test sets

To train the above networks, we used a training set that included 40 mosaics from the 11 Fiji Islands shown in Fig. 1. Each of the mosaics included one day of acquisition (WV-2 overpass).

One of the key requirements in developing accurate machine learning algorithms is an abundance of ground-truth data. However, it is well documented (e.g. Kemker et al., 2018; Maggiori et al., 2017) that there is a scarcity of labeled data for remote sensing applications, especially for multi-spectral imagery. Hence, when building a new CNN (or other machine-learning-based) model application, often the first step is to create the so called "ground-truth" dataset. Although there are some ground truth datasets that include clouds, for Landsat purposes (Hughes and Hayes, 2014; Zi et al., 2018), these tend to be limited in size and coverage, as they are also manually labeled by human operators. Furthermore, there are no such available products for the finer

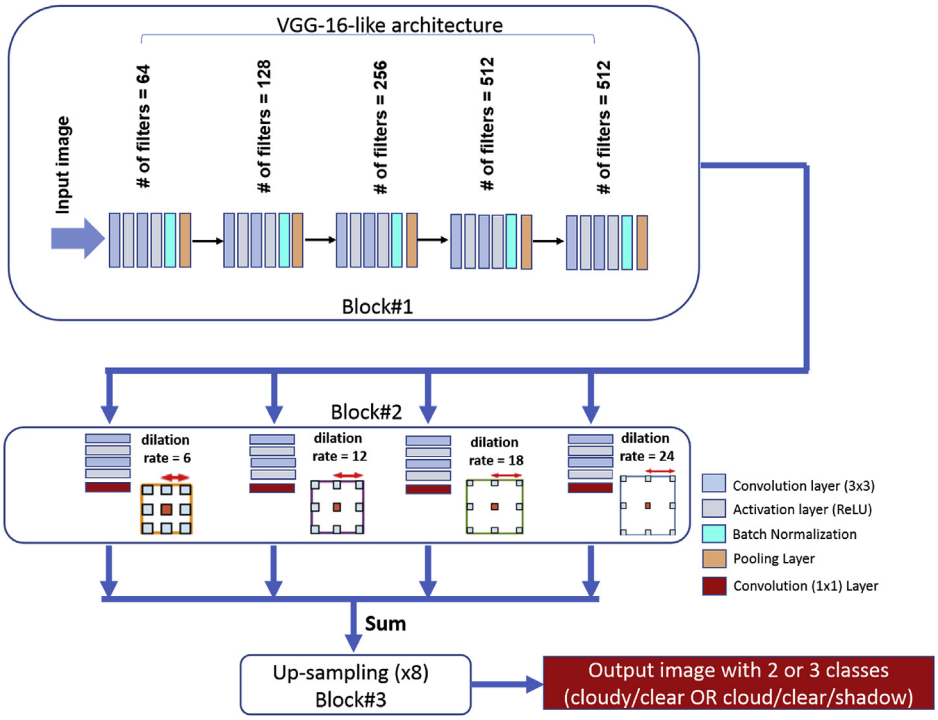


Fig. 5. An illustration of the DeepLab-like architecture (Chen et al., 2017) CNN scheme that was used in our cloud and cloud shadow detection algorithm. The ReLU activation layer is a Rectified Linear Unit and is a non-linear operation, which takes the max between zero and the pixel value. The network output results in a fully semantic segmented imagery with the same dimension as the input imagery (a 4-channel image patch). The advantage of this scheme is in Block#2, which uses sparse convolutions that can span various spatial coverage of the image patch, but without increasing computational burden much. Up-sampling is needed to reshape the downscaled output from Block#2 back to the initial input size. This same scheme was used for either a 2-class (clear/cloud) mask and for a 3-class (clear/cloud/cloud shadow) classification task.

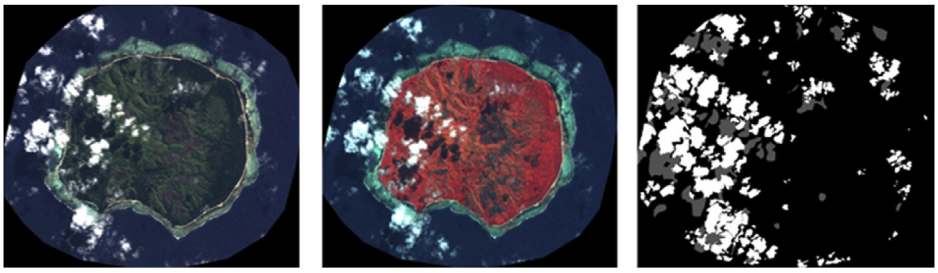


Fig. 6. (left panel) composite RGB image used as the basis for the cloud and cloud shadows labeling, (middle panel) false RGB image, where red channel is replaced by NIR channel, and (right panel) the manually classified clouds (in white), shadows (in grey), and all other cloud-free areas (in black). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

resolution platforms such as Sentinel-2 or WV-2, nor for other high-resolution platforms such as the Chinese satellites GaoFen-1 and GaoFen-2 (Chen et al., 2018). Hence, the creation of a manually-generated ground-truth dataset is the current standard for testing purposes of CNN applications in land and cloud classification tasks (e.g. Hughes and Hayes, 2014; Chen et al., 2018). For our purpose of devising cloud and cloud shadow masks, we manually annotated all 40 Fiji Islands mosaics from WV-2 using photoshop software, basing our annotations on RGB image composites, as shown in Fig. 6 (left panel), where clouds (both thick and thin) were labeled together (shown in white on Fig. 6 right panel), while shadows (which can be better observed using a false RGB image where the NIR channel is used instead of the Red channel, as in Fig. 6 middle panel) and cloud-free (which include all other areas in the image) were labeled seperately in grey and black, respectively (Fig. 6 right panel). Fig. 6 shows only one mosaic for the island of Cicia, but all other mosaics were labeled similarly.

The manually labeled cloud and cloud shadow masks were regarded as our ground-truth for the training and test purposes. Indeed, this might create difficulty in generating an absolute comparison between our approach and others. Hence, to compensate for these discrepancies, we have compared our selected working architecture (DeepLab) with two bench-mark algorithms, namely the MLP-NN and the VGG-16 architectures, as detailed above. 8000 256 × 256 × 4 image patches per class were used for training whereas 800 image patches per class were used for cross-validation/test through the training process. Image patches were selected randomly from each of the mosaics, with 200 (or 20) taken from each mosaic for creating the training set (or validation/test sets). The decision to use only 4-channels in our cloud detection algorithm was made after testing the algorithm with the 8-channels available by WV-2, which showed inferior results. This can be explained by the fact that the additional channels of WV-2 include more contamination from atmospheric trace gases such as ozone (Yellow band-585–625 nm), and water vapor (NIR2 band- 860–1040 nm), which we have not corrected for in our pre-processing procedure. Hence, the final selection of the bands shown in Table 1 was set to avoid this problem and to find the best match with corresponding Sentinel-2 bands that can be used for the application of the same cloud detection algorithm.

The training of the network was done using the Keras API python package (Chollet, 2015) using the TensorFlow (a system for large scale machine learning) backend (Martin et al., 2016). The training process was done in a mini-batch mode, with a batch size of 6 image patches of 256 × 256 pixels in each batch. Although processing was done with NVIDIA Cuda GPU, we could not increase the batch size due to computational resource limitations (12GB GPU RAM). Standardization (mean subtraction and standard deviation division) was applied on the input tensor (with dimensions of 256 × 256 × 4), with mean and std values of 100, which was derived from the entire imagery dataset. This yielded values in the range generally between −1 and 1. To augment the number of training samples, we applied, after the standardization process, a channel shift procedure, where values in the range of 0.1–0.25 were randomly added or subtracted to/from each of the channels. This was performed to try and simulate the slight shifts in radiance values between the scenes of WV-2, but also to simulate the shifted values between the WV-2 and Sentinel pixel intensities for

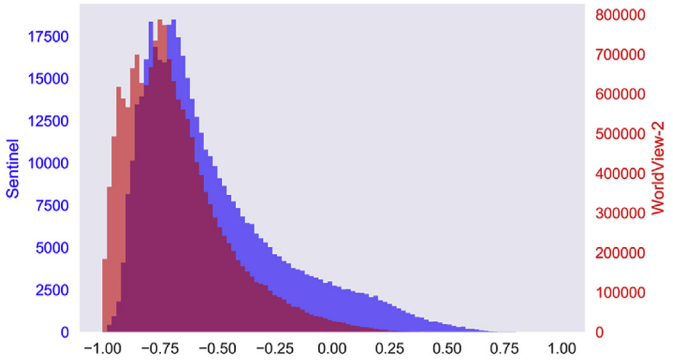


Fig. 7. Normalized (−1 to 1) pixel intensity frequencies in cloudy regions, compares for WV-2 (light orange) and Sentinel (light purple) averaged over the four channels shown in Table 1. Dark pink area shows the overlap of the regions. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

clouds, as seen in Fig. 7, which shows normalized pixel intensity values for both WV-2 (light orange), and Sentinel (light purple). The dark pink area shows overlapping pixel intensities and we see that the average shift over the four channels is between 0.1 (lower intensity range) to ~0.25 (higher intensity range).

During the training process, we used the Adam stochastic gradient-based optimizer, which is a first-order gradient-based optimization of stochastic object functions (Kingma and Ba, 2014), with initial learning rate of 1E-4, and a weight decay rate of 3xE-3 to regularize the training process. With the Keras API, this regularization is implemented (multiply the total loss) on the final loss function rather than on the network weights. We have trained the network from scratch (i.e. not using previously trained weights for the VGG-16 CNN scheme), and used a softmax activation as our output layer (which outputs a probability distribution for each class) with cross-entropy as our loss function. Cross-entropy loss, or log loss, measures the performance of a classification model whose output is a probability value between 0 and 1. Cross-entropy loss increases as the predicted probability diverges from the actual label. So, a perfect model would have a log loss of 0:

$$Loss = \sum_{c=1}^{M} y_{o,c} \log(p_{o,c}) \tag{3}$$

where M indicates number of classes, y is a binary indicator (0 or 1) if class label c is the correct classification for observation o, and p is the predicted probability of observation o is of class c.

As cross-entropy indicates the distance between what the model believes the output distribution to be and what the actual distribution is, this function is widely used as an alternative to the square error loss in classification problems. Finally, we trained over 1000 epochs with 1000 steps per epoch, which took about four days on our hardware.

2.2. Domain adaptation neural-network

Training a new CNN (or any machine-learning-based algorithm) network for each satellite platform is very costly, since a separate network model, using a separate ground-truth set (generated manually) must exist. As will be discussed next, a network trained for one platform might not always perform optimally on another, and often has to be manually tweaked or modified to perform well, constraining such algorithms from wide-spread operational usage. Similar issues also exist with rule-based algorithms such as the Landsat Fmask (Zhu and Woodcock, 2012), where a new set of threshold has to be established per each rule per each platform. In the latter case it was shown that the Landsat based Fmask algorithm did not perform ideally when applied on Sentinel-2 imagery (Zhu and Woodcock, 2012). Here, we introduce a method that has a potential to overcome this issue and to allow the classification of cloud-free and cloudy pixels for different multi-channel satellite platforms. Specifically, we have focused on a scheme that can learn from both WV-2 and Sentinel-2 platforms, leveraging the fact that we can use the same set of inputs (4 channels of RGB + NIR) for both satellites, although this approach can be expanded to accommodate other platforms as well, using similar channels, such as Landsat. The suggested scheme is a domain adversarial NN (DANN) developed by Ganin et al. (2016), which has shown promise in classification problems. This approach is inspired by the theory of domain adaptation, which claims that predictions using inputs from different sources can be made based on the features that are invariant between such inputs, for example edge/shape features or spectral ratios (rather than fixed ones). The DANN approach is often used in cases where there exist a ground-truth/labeled dataset for only one of the input sources. Hence, its goal is to be able to perform a prediction on the input lacking the appropriate labels using the information from the labeled input dataset. In the DANN nomenclature the labeled input is named the source (WV-2 in our case) and the un-labeled input is named the target (Sentinel-2 in our case). In essence, by utilizing the DANN approach one might cut in half the time needed for the training and truth labeling when generating a classification model for two different satellite platforms. This is since the DANN scheme uses only the 'source' truth labels during the training process, so there is no need for the generation of a new truth set for the 'target' platform. Nevertheless, the cloud detection case is unique in that sense since clouds are dynamic and cannot be reproduced between different scenes that are captured among different satellites (and also among the same satellite but at different times). Hence, in order to calculate our prediction accuracy on the 'target' platform during test time there was a need to develop a labeled truth set for the 'target' domain. This is un-necessary when performing the same task on marine habitat classification, for example, since the truth-labels (such as corals, sand, sea-grass, ocean etc.) are considered static within the different scenes captured. The network architecture and approach used in this work is shown in Fig. 8. In general, this is a relatively simplified network, which contains a feature extraction block, with two convolution and pooling layers, similar to Block#1 shown in Fig. 5, but shallower in nature. Then, the network partitions into two branches, where each branch accomplishes a different role. As shown in Fig. 8, the right (classification/label predictor) branch is a multilayer perceptron, which classifies either clear or cloudy pixels using the softmax activation layer with cross-entropy loss as described in the former section. This architecture was chosen to simplify the training process and to show the proof-of-concept that such an approach might be useful for such a task of cloud classification from different platforms.

The left branch (discriminator/domain predictor) performs the task of learning a discriminative classifier between the source (WV-2) and target (Sentinel-2) distributions. During the training process, images from both WV-2 and S-2 are used as inputs, but only ground-truth labels for WV-2 are used in the minimization of the classification loss. In parallel, the domain discriminator branch attempts to maximize the loss and forces a random discrimination between the two domains so that during test time (when the domain discriminator branch is not active), the network can perform interchangeably on inputs from either platform (the optimal accuracy of the domain discriminator branch should be approximately 0.5). The key aspect to this branch is the gradient reversal layer which ensures that the feature distribution over the two domains are made similar (as indistinguishable as possible for the domain classifier), and that the original feature extractor will instead converge on domain-invariant features. Such domain invariant feature can be edge/shape of the clouds apparent in both sensors, or the relative intensity of the cloudy versus the cloud-free/clear pixels rather than their absolute value. The overall network loss function minimizes the sum of the losses from each of the branches (label prediction plus domain discriminator).

During the network training, we proceed as if we do not have any S-2 labeled data and introduce only our labeled data for WV-2. We do

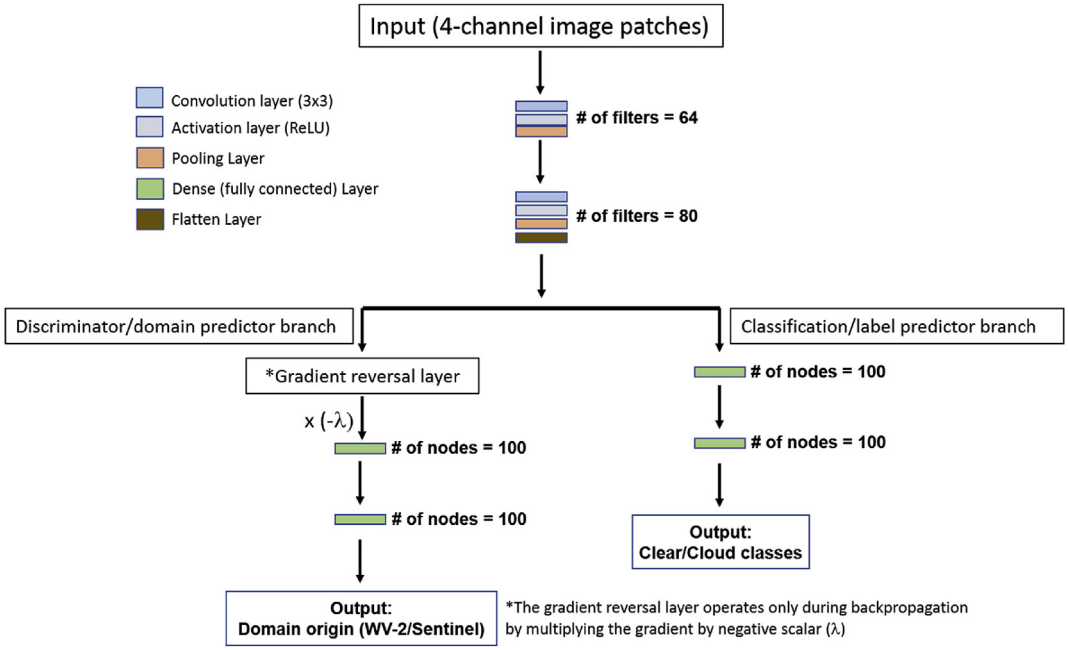


Fig. 8. An illustration of the DANN network architecture used in our retrieval algorithm flow for clear/cloud classification for either WV-2 or Sentinel images.

introduce, however, S-2 labeled data for the accuracy testing of the network predictions, as detailed below.

To test the DANN approach we have trained a network as shown in Fig. 8 where we used as input 3600 training samples of size $80 \times 80 \times 4$ per each class from 18 different WV-2 mosaics (i.e. 'source' input) and additional 1800 training samples of size $80 \times 80 \times 4$ per each class from 9 different S-2 mosaics (i.e. 'target' input). For testing purposes, we used 360 randomly selected $80 \times 80 \times 4$ WV-2 patches and 180 S-2 patches, where the latter had to be manually labeled in order for us to generate our predicted S-2 accuracy values. One difference that was introduced in this framework, in compare to our augmented CNN approach above, is that the trained DANN uses decimated WV-2 images that match the Sentinel-2 resolution. This was done to allow the evaluation of the DANN on the spectral domain differences rather than on both the spectral and spatial ones. To test how much the domain adaptation approach might improve the classification of the Sentinel-2 data versus the baseline scenario, we trained the overall network twice: once, using only WV-2 as our source domain (with labeled images), without the utilization of the domain predictor branch, hereafter referred to as the 'source' training, and a second time using both WV-2 and Sentinel-2 as inputs (S-2 without labeled images), hereafter referred to as 'dann' training. During the 'dann' training, the network operates with both branches (classifier and domain discriminator), to build the mapping between the two domains, so that the classifier is able to extract domain-invariant feature across both source and target domains. We used 10 epochs, with 10000 steps per epoch, a mini-batch size of 64, and an adjustable learning rate based on Ganin et al. (2016).

During test time (prediction) only the right branch (classifier) of the network is active as it acts as the optimized label classifier for both domains (either WV-2 or Sentinel-2, depending on which domain we are testing). As mentioned above, to be able to calculate prediction accuracy of the S-2 imagery, we have generated manually labeled truth data for this testing step.

3. Results

This section summarizes our results as follows: (1) Cloud mask CNN (various architectures) applied to WV-2 imagery, (2) cloud and cloud shadow masks CNN (selected architecture) applied to WV-2 imagery, (3) augmented cloud mask CNN (selected architecture) applied to Sentinel imagery, and (4) cloud mask DANN framework applied to Sentinel imagery.

3.1. Accuracy of CNN retrieval algorithms

For only cloud masking, we have compared the three architecture types, as described in section 3.1.1. The purpose of this comparison was to establish the accuracy results of the DeepLab architecture for our ground-truth WV-2 dataset with two bench-mark architectures, namely the MLP-NN (standard NN) and the VGG-16 (standard CNN). We calculated accuracy based on predictions made on image patches of the size of 2048×2048 pixels from multiple mosaics, comparing predicted values to our manually labeled 'ground-truth' data. The prediction process was processed in a sliding window mode, predicting on patches of 256×256, with an overlap of 1 pixel for the MLP NN and VGG-16 models (essentially predicting one pixel at a time), and 128 pixels for the DeepLab model. For the combined cloud and shadow masks the accuracy metrices were calculated based on predictions on entire mosaics, as the number of shadow pixels is overall low.

The following metrices are used in our accuracy evaluation of the various networks:

Overall network accuracy (Acc) is calculated as the sum of hits (correct classification) from each of the classes divided by the sum of total pixels to classify. This measure is biased towards the class with the greatest number of pixels, which is usually the clear (cloud-free) pixel class, and will often overestimate the network accuracy. Hence, the metrices below were also calculated to examine each class separately.

Probability of detection (POD), also referred to as hit rate, is calculated as the number of hits (correct class) divided by number of hits plus number of misses in each of the classes, and is calculated per each class separately.bullet

Omission error (OME), also referred to as miss rate (correct class instances that were omitted/left out from the correct classification), is calculated as the number of misses (e.g. clouds classified as cloud-free pixels) divided by the total number of pixels in a given class.-bullet

False Alarm Ratio (FAR), which can also be referred to as

Table 2
Cloud detection statistics compared between the three network architectures (in percentages), based on the results shown in the confusion metrices in Fig. 9

	Overall Acc	POD	OME	FAR	CSI
DeepLab	94	89	11	16	75.6
VGG-16	54.7	92.9	7.1	70.7	28.7
MLP NN	84.5	60	40	35.5	45

commission error, is calculated as the total number of false detections (e.g. classified as clouds but are actually cloud-free) divided by the number of hits (e.g. for clouds) plus the number of false detections.bullet
Critical Success Index (CSI) is calculated as number of hits in each class, divided by the number of hits plus the number of false detections plus the number of missed detections (miss rate).bullet

3.1.1. Clouds only

In the cloud only classification, the networks had the binary task of predicting either cloudy or cloud-free pixels. Table 2 compares Acc, POD, OME, FAR, and CSI as detailed above for the cloud class, while Fig. 9 presents the confusion matrices associated with each of these architectures for both classes. In the figure, each 2 × 2 matrix shows the overall class statistics along the margins, where along the rows, the accuracy for each class is shown as percentages (in green hues). The MLP-NN and the DeepLab networks overall accuracy is high with 94%, and 84.5%, respectively, where the overall accuracy for the VGG-16 network is ~55%. Although low, the latter is doing well in detecting clouds but underpredicts the clear pixels. This is in contrast to the MLP-NN scheme, which shows inferior prediction capability for the cloudy pixels. The three networks differ in their POD level of the cloudy pixels, with 60%, 93% and 89%, for MLP-NN, VGG-16 and DeepLab, respectively. Although the VGG-16 seems to have the highest POD for the cloudy class, the DeepLab architecture has high prediction capability and low OME values for both classes, as well as the highest CSI value for clouds. This translates to the fact that this architecture excels over the other two in detecting the cloud features within the image patch, incorporating more of the context of the image. Specifically, we have noticed (not shown) that the MLP NN tended to predict coastal

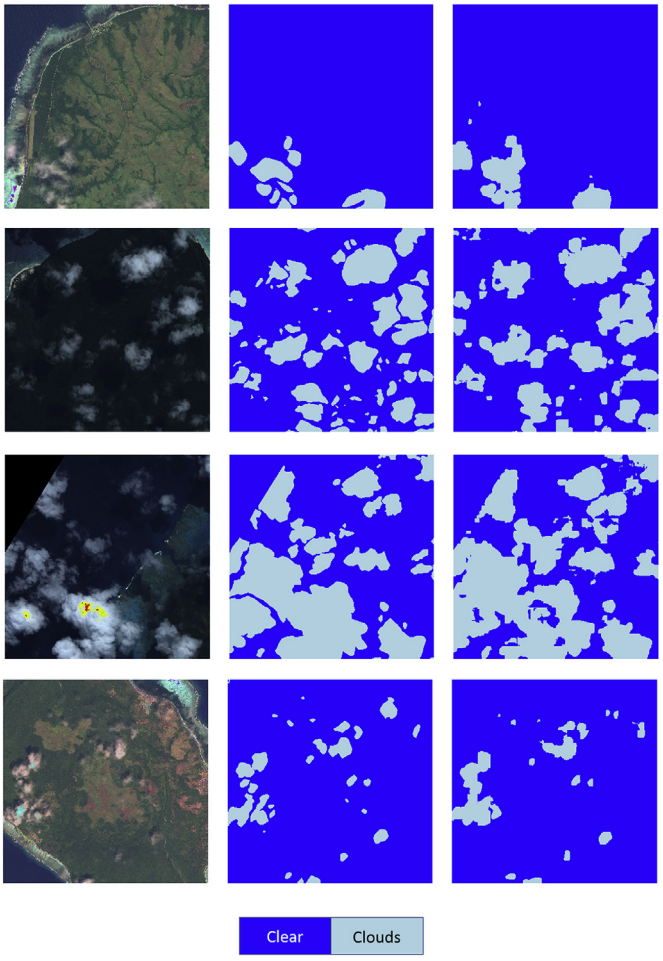


Fig. 10. (left panel) RGB patch image used in the test prediction, (middle panel) ground truth clear/cloud mask, and (right panel) clear/clouds labels predicted by the DeepLab architecture.

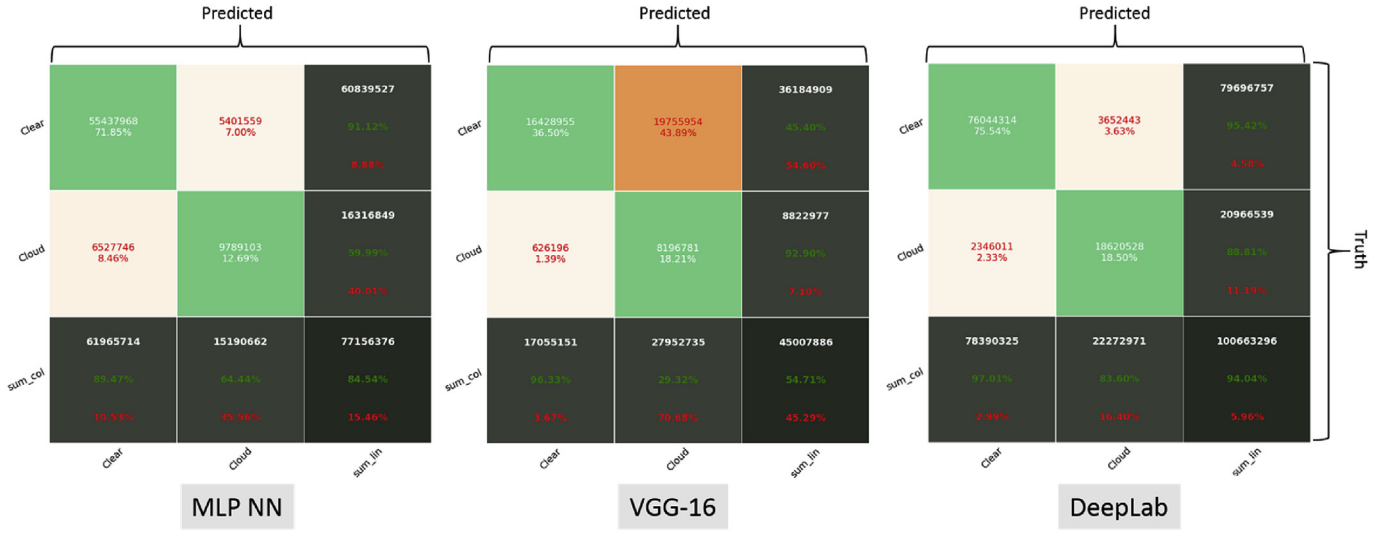


Fig. 9. Confusion matrices for clear(cloud-free) and cloud predictions by a traditional MLP NN architecture, similar to the one shown in Fig. 3, but with four hidden layers, as detailed in section 3.1.1, a VGG-16 CNN image classification architecture, as detailed in text, and DeepLab CNN architecture, similar to the scheme shown in Fig. 5. Green boxes represent the number of pixels which were correctly classified for each of the classes, light orange represent false classifications, and the dark green boxes along the margins of the matrix summarize the various hit/miss ratios (in percentages), where hit ratios are shown in green and miss ratios in red. Total number of pixels in each class and the sum of classified pixels are shown as absolute numbers, where the relative amount of each class is shown in the inner 2 × 2 matrix as percentages. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

(brighter) areas as clouds, probably due to their relatively high intensity that was assigned by the network to the cloud class. The low prediction capability for the cloud-free pixels by the VGG-16 network are surprising but this might stem from the fact that the cloud-free class included multiple other classes (i.e. clear sky, including both ocean and land), which could not be optimized using the single pixel classification approach. With the DeepLab architecture results showing superiority to that of the two bench-mark architecture tested, our ensuing examples, analysis and discussion will refer to this architecture.

Fig. 10 shows some examples of the predicted patches by the DeepLab architecture (hereafter denoted as the selected architecture), where the left panel displays the RGB image, middle panel the manual labeled ground truth and right panel the predicted cloud mask. As shown, clouds are predicted fairly accurately over both land (all four rows) and water (third row). In the first row we see that some clouds are overpredicted over narrow valleys over land, where no clouds are observed in the original image or the ground-truth patch. In addition, it seems that the predicted pixels (right panel) encompass larger areas when compared to the ground truth. This is possibly due to the upsampling (via bilinear interpolation) effect in the DeepLab architecture where neighboring pixels are being inferred (interpolated) in order to scale back the downsampled patch into its original input size.

3.1.2. Clouds and cloud shadows

For this prediction task, we attempt to classify clear (cloud-free), cloudy and cloud shadow pixels using the DeepLab architecture. The prediction was processed in a sliding window mode, predicting on patches of 256 × 256, with an overlap of 128 pixels, and was performed for 15 whole mosaic patches. Entire mosaics (rather than 2048 × 2048 patches) were used here since the relative amount of cloud shadow pixels is ~5% of the overall pixels (Fig. 11, third row of

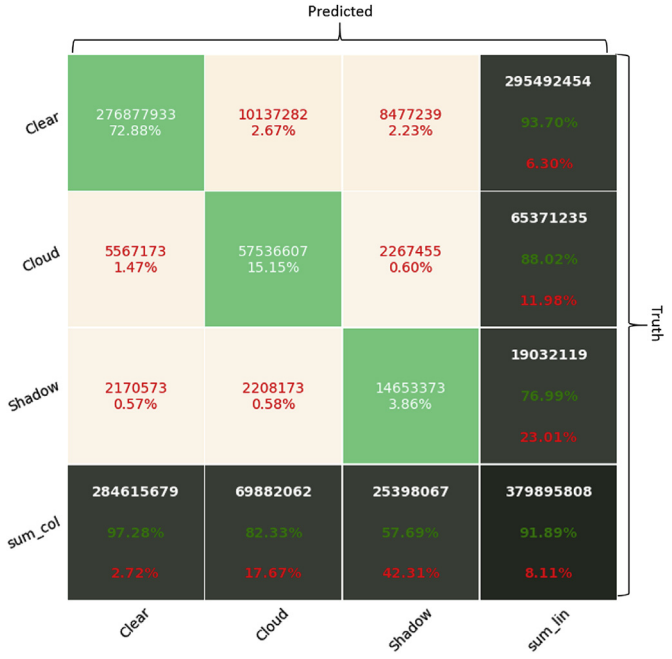


Fig. 11. Confusion matrix, similar to Fig. 9, but for clear (cloud-free), cloud and cloud shadow predictions. Green boxes represent the number of pixels which were correctly classified for each of the classes, light orange represent false classifications, and the dark green boxes along the margins of the matrix summarize the various hit/miss ratios (in percentages), where hit ratios are shown in green and miss ratios in red. Total number of pixels in each class and the sum of classified pixels are shown as absolute numbers, where the relative amount of each class is shown in the inner 2 × 2 matrix as percentages. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

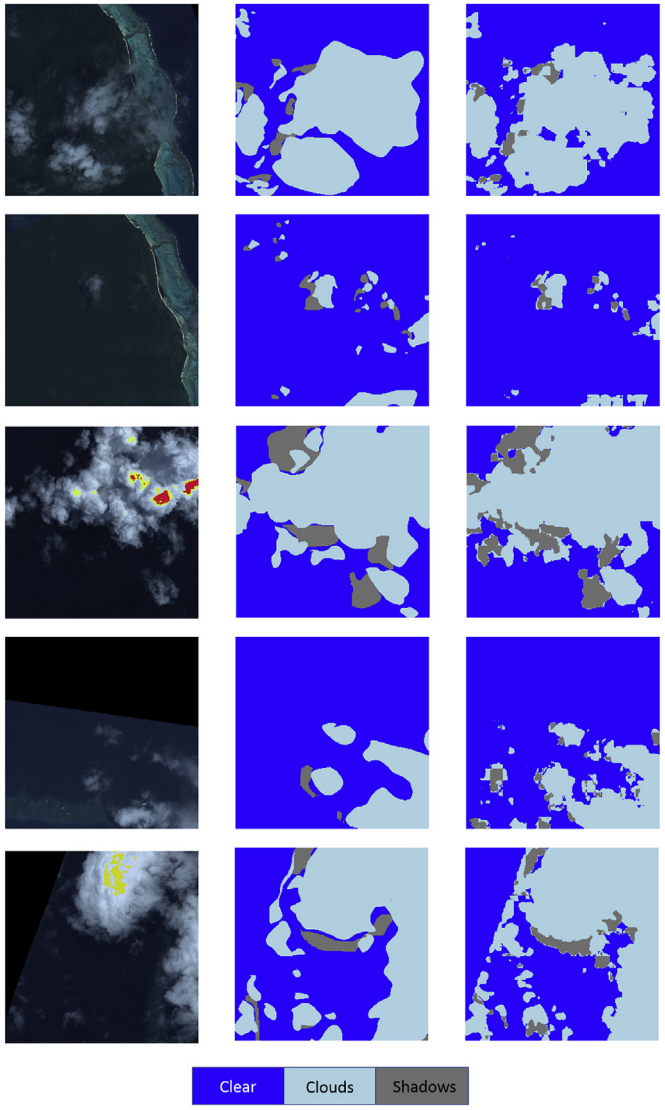


Fig. 12. (left panel) RGB patch image used in the test prediction, (middle panel) ground truth clear/cloud mask, and (right panel) clear/clouds/cloud shadows labels predicted by the DeepLab algorithm.

the confusion matrix). From Fig. 11 we can see that in comparison with the cloud-free vs. cloud binary classification, the overall accuracy of this algorithm is 92%, which is slightly reduced from the cloud-free/cloud only DeepLab network (with 94%). This seems mainly due to the lower POD value of the cloud shadow class. Nevertheless, due to the fact that cloud shadow pixel amount is relatively low, this does not greatly affect the overall accuracy. FAR for the cloud-free and cloud classes is very similar to the cloud-free/cloud only network (~3/17%, respectively), but much higher for the cloud shadows (~42%), with mostly cloud-free pixels classified as cloud shadows, which occurs especially over dark surfaces. OME for the cloud shadows is divided almost equally among the cloud-free and cloudy pixels. Nevertheless, we see (Fig. 12 and Fig. 13) that the algorithm manages to identify both cloud shadows that are adjacent to clouds and those that are further apart. For the clouds class, almost 12% of the pixels are falsely classified as either cloud-free or shadows. The majority of the shadow mis-classifications seem to occur in transition regions on cloud edges, where the cloud is thinner and usually overlays a dark surface.

Fig. 12 is similar to Fig. 10 but showing original patch RGB image (left column), ground-truth labels (middle column) and prediction labels (right column) for the clear/cloudy/cloud shadows classes. The

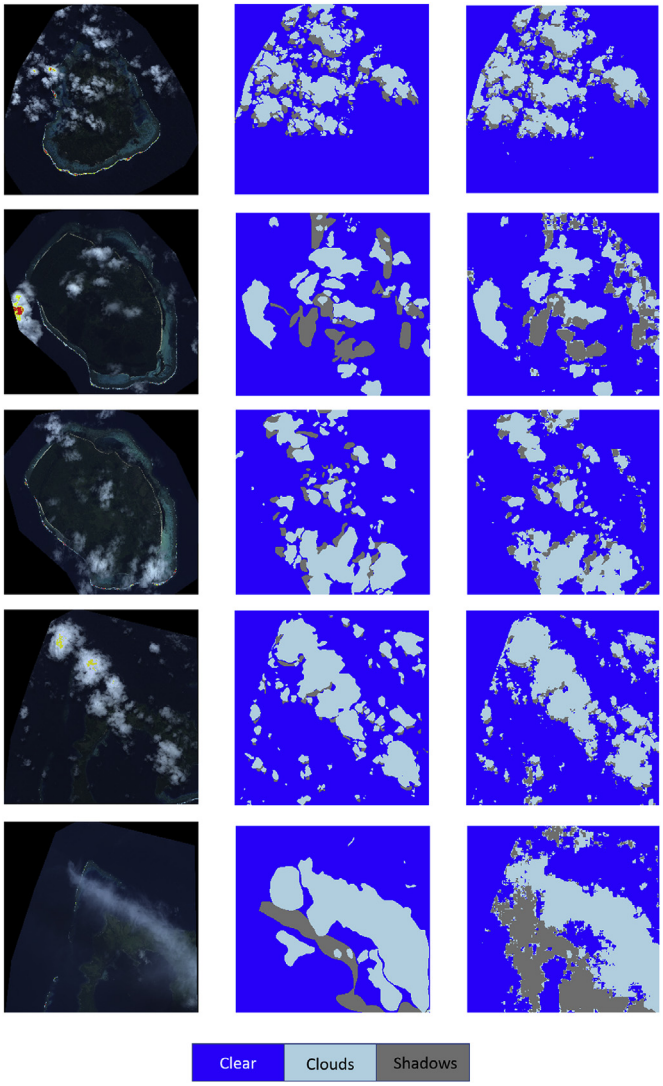


Fig. 13. (left panel) RGB island mosaic image used in the test prediction by the DeepLab scheme, (middle panel) ground truth clear/cloud mask, and (right panel) clear/clouds/cloud shadows labels predicted by the multi-class algorithm, for Matuka (1st row), Nayau (2nd and 3rd), and Totoya (4th and 5th) islands.

first and second rows represent cases where cloud shadow POD rates were low, with 70% and 40% for the first and second rows, respectively. As seen from the examples, these low rates seem to be a result of a very complicated scene (cloud shadows are very adjacent to and sometime mixed with the cloud scene), with some uncertainties in the ground-truth map itself. This might also be due to the smoothing that happens during the upsampling procedure, or the classification itself, where the boundaries between cloudy and cloud shadow pixels are not well separated. The third row in Fig. 12 represents a case where 15% of the true cloud-free pixels were classified as clouds. This, again, is attributed to the smoothing effect of the upsampling operation and also to the conservative ground-truth cloud labeling, which resulted in classifying even the patchy, semi-transparent (i.e. mixed cloud-free and cloudy pixels) as clouds. In the fourth row we see an example of overprediction of cloud shadows, with the rate of 50%. This is possibly due to the sub-optimal color scheme of the imagery, which made the algorithm classify the relatively grey-shade cloud-free areas as shadows. The last row in Fig. 12 represents a case where cloudy pixels are falsely classified as cloud shadows. We see here that the classification scheme confuses semi-transparent clouds (over the dark ocean) as shadows. In addition, we see that most of the cloud shadow misclassifications (either between cloudy or cloud-free pixels) occur over the dark ocean, where cloud shadows are hard to distinguish from the dark background. Nevertheless, as our algorithm uses both the spectral and the spatial information for inference, we do not observe misclassifications of wave-breaks (which are very bright) as clouds (Fig. 12, first and second rows), as might happen with threshold-based algorithms. Finally, the overall performance of the multi-class algorithm seems to be good, and we note that many of the aforementioned misclassification examples are very hard to identify even to the human eye.

3.2. Sensitivity of results to cloud and surface types

We have examined further our multi-class classification (cloud-free, clouds, shadows) in terms of prediction of clouds and cloud shadows over ocean or land and the prediction of thin clouds such as cirrus type clouds. Shown in Fig. 13 are some classification examples over entire island mosaics that demonstrate our algorithm sensitivity to the above.

As seen from Fig. 13, the algorithm performs well in both cloud and cloud shadow predictions overall. Specifically, in the first row we see prediction over Matuka island, where well-structured puffy cumulus clouds are present over both land and ocean. Here, the cloud-free and cloud classes are predicted with 95% accuracy, whereas the cloud shadows with 80%. Comparing the next two rows for Nayau island, we see that here we have more cloud shadows over ocean, which is predicted overall with better accuracy than cloud shadows over land (as seen from the second and third rows). Specifically, with the latter two, cloud shadows are overpredicted instead of cloud-free pixels, especially over land, with a FAR of about 17% for both mosaics on the second and third row of Fig. 13. On the third row of the figure (right-most column) we see a cloud shadow mis-classification over the dark coastal patch. We believe that this arises from both the similar dark spectral thresholds, but also from the unique shape of this coastal patch, which was mistaken to be a cloud shadow. Nevertheless, the cloud shadow prediction is performing well for both cloud shadows that are adjacent to the clouds (1st and 3rd rows), as well as for cloud shadows that are more distant (2nd row). The fifth row shows an example of a different type of cloud, which is thin and wispy (a cirrus cloud). These types of clouds were rare in our training set, with the majority of the cloud types identified as cumulus clouds. In this case the cloud POD for this mosaic is 71%, where the mean POD is 89%. The main issue here is that clouds were wrongly predicted either as cloud-free pixels (11%) or cloud shadow pixels (17%). This is because the cloud is very thin and the underlying surface plays a larger role.

3.3. WV-2 trained CNN algorithm applied to Sentinel-2 imagery

In this section we present prediction results using our DeepLab CNN network architecture, trained only with WV-2 images, on Sentinel-2 images taken from the Fiji Islands. As mentioned above, we have trained our network with a normalized channel shift augmentation of 0.1–0.25, where results from the 0.25 augmentation are shown here. To compare the performance of our augmented WV-2 based CNN network with the original Sentinel cloud mask, we had to generate our own ground truth set. We have generated a manually labeled truth cloud mask for randomly selected 18 images of Sentinel from Cicia, Fuluga, Kobara and Mago islands, spanning a date range between January 2016 to December 2017. We applied our CNN prediction on 19 randomly selected 512×512 pixel patches generated from Sentinel imagery. Since our original network was trained using only WV-2 data, which had different mean and standard deviation values than the Sentinel images, we must normalize the Sentinel imagery before introducing them to the network for prediction. Mean and standard deviation values were calculated across the four input channels and were used to normalize each input patch used for prediction. The results shown here used normalized input patches that were subtracted by two times the

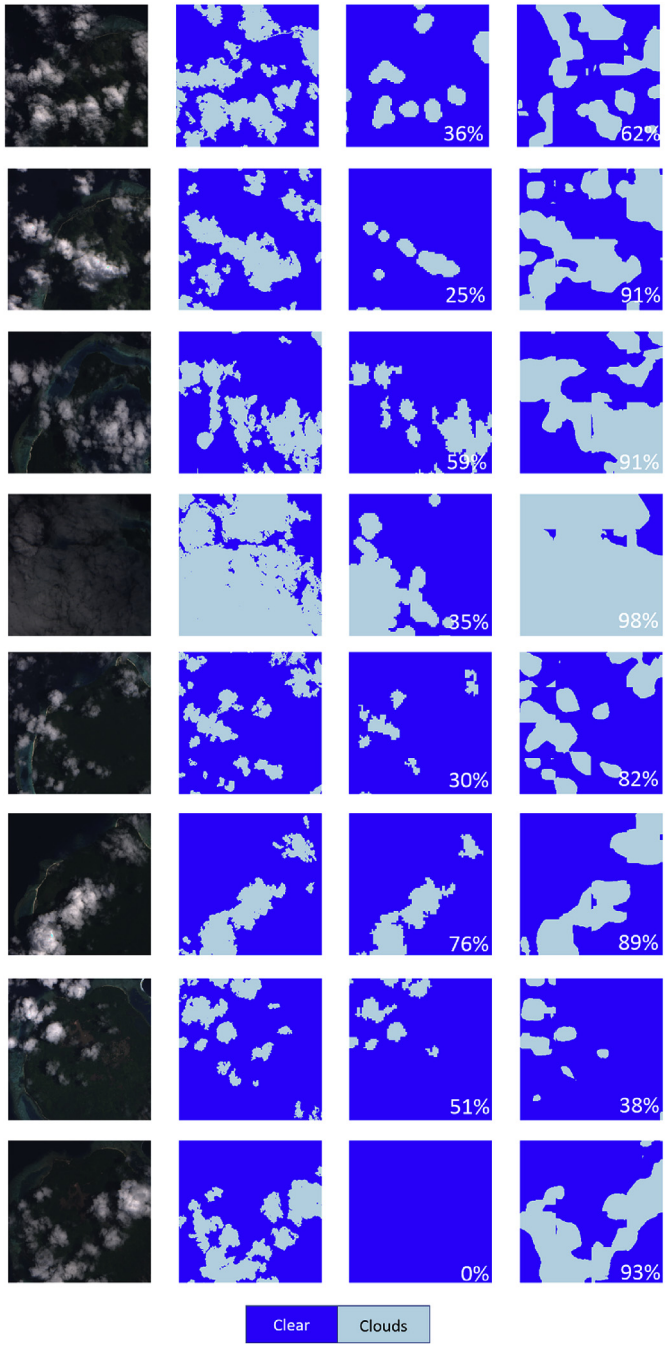


Fig. 14. (left column) Sentinel RGB island patch image used in the prediction by the WV-2 CNN based algorithm, (second column) manual generated ground truth cloud-free/cloud mask, (third column) Sentinel cloud mask prediction (with cloud POD in percentage), and (fourth column) WV-2 CNN based cloud mask prediction (with cloud POD in percentage).

overall channels mean and divided by two times the overall channel standard deviation. We have tested several such pre-process normalization combinations (e.g. subtraction by one mean and dividing by one standard deviation, or subtraction by two times mean and division by one standard deviation), and noticed that the network prediction results are sensitive to this initial normalization. Note that normalization itself is only a linear transfer between the two spectral spaces, and cannot capture nonlinear effects that will naturally occur when moving across both datasets. Fig. 14 shows some examples of the original Sentinel image patches as RGB images (left column), our own manually generated cloud truth mask (second column), the original Sentinel cloud mask prediction (third column), and the prediction by our cloud mask CNN trained only with WV-2 data, with a 0.25 channel shift augmentation. Each row represents a different patch (different island and different dates), with the percentages of the cloud POD for the original Sentinel cloud mask algorithm (third column), and our WV-2 based CNN (fourth row) shown on the figure. As seen, both cloud mask schemes are not perfect, but in general, the original Sentinel cloud mask highly underestimates the amount of clouds (predicting clear pixels where clouds are present), while the WV-2 based cloud mask often overestimates the cloudy sections (predicts clouds even on cloud-free regions). The latter is consistent with what we have seen earlier, where our CNN algorithm tends to smooth and increase areas predicted as clouds, due to the upsampling procedure and also due to the relatively conservative training set used for the WV-2 CNN network (i.e., even relatively see-through cloud patches are classified as clouds). As seen in Fig. 15, overall accuracies (including cloud-free and cloudy prediction rates) for each of the schemes are 77% (left panel confusion matrix) and 82% (right panel confusion matrix) for the Sentinel and WV-2 based schemes, respectively. While these seem very close, they are substantially different, since the Sentinel cloud mask excels more in predicting cloud-free pixels (~92%) versus only 81% for the WV-2 based CNN scheme, and vice versa for the cloud POD, with 48% and 84%, respectively. Fig. 14 shows relatively higher POD values (see percentages on the lower right side of the images in columns three and four of the figure) for most of the WV-2 CNN based cloudy pixel predictions. Lower WV-2 prediction rates are seen for cloud-free pixels. It is interesting to note that the lowest classification rates by Sentinel (rows 2,4,5, and 8) are for regions where the background seem darker, as well as the clouds. Fig. 15 follows Figs. 9 and 11 showing the mean confusion matrices derived based on our 19 patch predictions, where left panel shows prediction of the Sentinel cloud mask relative to our manually labeled ground-truth and the right panel shows WV-2 based cloud mask predictions.

As seen from Fig. 15, Sentinel cloud mask shows prediction accuracy of 92% for the cloud-free pixels, and OME of about 8%. However, it has substantially low POD for cloudy pixels (48%), with OME of approximately 50%, which means that cloudy pixels are classified as cloud-free pixels about 50% of the times. Application of our WV-2 based augmented CNN algorithm on the Sentinel images gives a prediction accuracy of 81% for the cloud-free class, and 19% OME for cloud-free pixels. However, it has much lower OME (~17%), and higher POD (~84%) for the cloud class, compared with the Sentinel algorithm performance. Cloud prediction rates in Fig. 15 corroborate the examples shown in Fig. 14, where the original Sentinel cloud mask under-predicts clouds when there are clouds (POD of 48%). Also, we see here that the WV-2 based augmented CNN algorithm has higher cloud POD of 83%, albeit with higher FAR on the clouds category, where cloud-free is predicted as clouds about 30% of the times. This is higher than Sentinel cloud mask (25%) for the corresponding misclassification error. We have also noticed that the WV-2 based augmented CNN algorithm has higher standard deviation values over the patches tested here, with 18% and 24% standard deviations for cloud-free and cloudy predictions, respectively. This is expected, since, as we have mentioned earlier, the prediction was sensitive to the input patch normalization values used.

3.4. Application of DANN cloud mask retrieval to WV-2 and Sentinel-2 imagery

The prediction rate results for cloud-free and cloudy pixels based on the DANN network run, as described in section 3.2 are shown in Table 3. Avg. accuracy represents the average network accuracy including both classes, in percentages.

We define Source training (tr) as the training process when it proceeds only through the classification predictor branch (Fig. 8), while training on the WV-2 data as input. The DANN training is when the

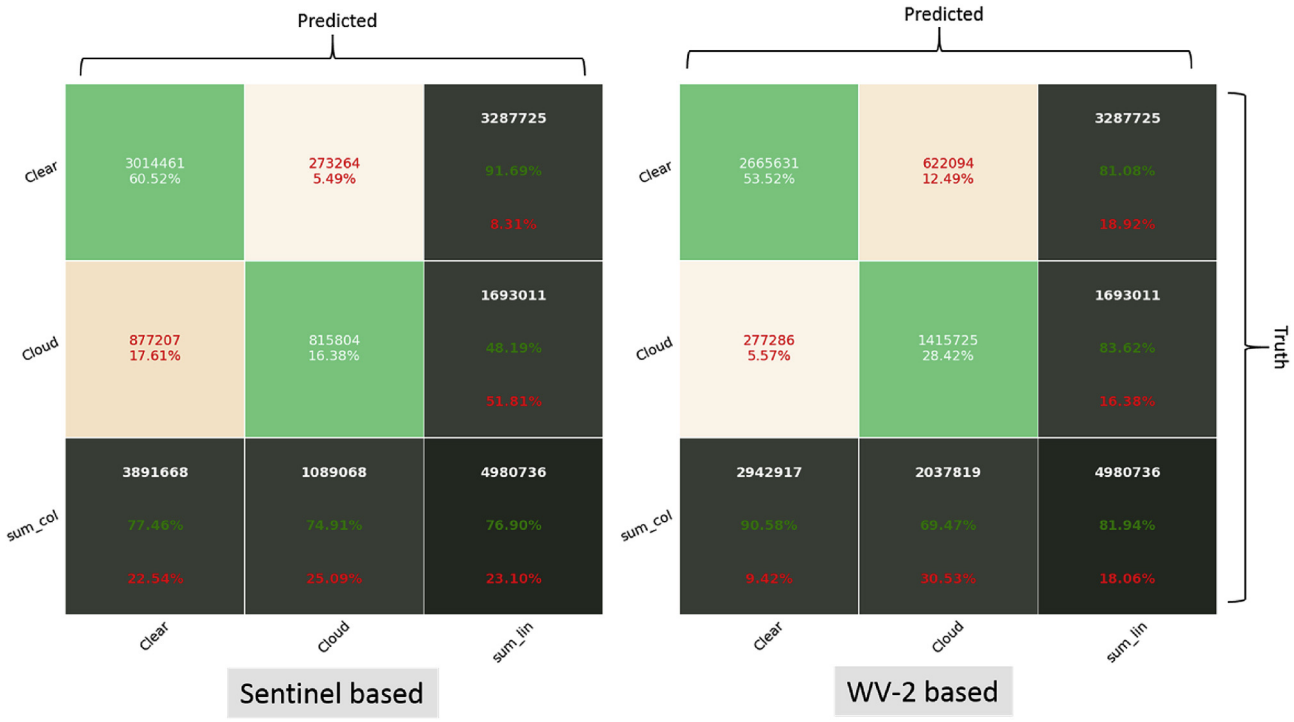


Fig. 15. Confusion matrix, similar to Fig. 9, for cloud-free/cloudy predictions, based on (left panel) Sentinel cloud algorithm (downloaded with the imagery), and (right panel) WV-2 DeepLab cloud algorithm.

Table 3
Accuracy rates (PODs in %) for DANN based cloud detection algorithm.

	Avg. accuracy	Clear	Clouds
Source tr, Source test	97	95.7	98.3
DANN tr, Source test	98.8	98.9	98.8
Source tr, Target test	78	64.5	91.6
DANN tr, Target test	84	73	95

training process proceeds in parallel over both branches (classification and domain discriminator), while minimizing both losses in the optimization process. This process uses both the WV-2 data and S-2 data as inputs, to learn the common features of each of the two domains. At test time, several prediction options were examined: Source test is when a patch from the source imagery (WV-2) is predicted by either one of the networks ('source' or 'dann' training), and Target test is when a patch from the target imagery (S-2) is predicted by either one of the networks ('source' or 'dann' training). From the table we see that for the source training only, the prediction accuracy of clear and cloudy pixels for 'source' test patches is very high (mean accuracy of 97%). This is even higher than the results obtained by the different CNN architectures as presented above. However, in this case, our results reflect the prediction on only 360 80 × 80 × 4 WV-2 random patches from 18 mosaics, which corresponds to prediction of 360 pixels (remember that this is a central pixel classification), as opposed to the prediction of full 2048 × 2048 × 4 patches from 24 mosaics, which translates into ~1E9 pixels. We also see that the prediction accuracy of clear and cloudy pixels for 'source' test patches using the 'dann' model gives even slightly higher accuracies for both classes. This might stem from the fact that the DANN approach is geared towards the optimization of specific features that are common to both domains, which seems to result in the selection of better cloud/clear features that might produce better prediction results. However, the more interesting results, and the ones relevant to our goal, are the Target test results for either the 'source' or 'dann' training schemes. Here, we see that the target test patches predicted (200 randomly selected patches of 80 × 80 × 4) by the 'source' model produce lower average and per class accuracies than the former ones ('source' predicted by either 'source' or 'dann' models), with the clear class resulting in lower prediction accuracy. After the DANN application, both classes prediction accuracies increase, with the clear (cloud-free) prediction accuracy increase of 12%.

Although not huge in terms of relative increase, we see that the DANN approach is capable in predicting cloudy and clear pixels at higher accuracy rates when compared to a simple application of 'source' only approach. In absolute terms, these cannot be compared on par with the former CNN results since the size of the training sets, test sets and the network architectures are different. We will continue to refine these comparisons in a follow-up manuscript where we apply the DANN approach on both the cloud/clear/cloud shadow classes, together with our marine habitat classes.

4. Discussion

4.1. CNN networks for predicting clouds and cloud shadows

We have presented several NN-based architecture schemes for cloud and cloud shadow detection. For cloud detection only we have compared two bench-mark methods, namely MLP-NN and VGG16-CNN networks with the DeepLab CNN architecture. For the combined cloud and cloud shadow detection we have used the latter architecture, as it generated results superior to the bench-mark algorithms when compared using the same ground truth test sets. Comparing our selected (DeepLab) cloud-only algorithm to existing state-of-the-art algorithms (in Foga et al., 2017) we see that we perform similarly to the best algorithm CFmask (with 90% correct rate values and 8% false rate values), which is a translation of the Fmask algorithm (Zhu and Woodcock, 2012) to C. Indeed, our algorithm was not trained or tested on a global dataset, but we observed that for a cloud-only mask we achieved high POD, with relatively small overall network FAR of 5.9% and a standard deviation among the sampled patches of 10%. When comparing our cirrus (thin wispy high-altitude cloud layers) prediction rate, as shown in Fig. 13 last row, we got 71% versus 85% for the

CFmask algorithm. However, most of the cloud types observed over the Fiji Islands were of the cumulus type (puffy cloudy patches), with occasional marine stratocumulus (a more homogeneous cover/blanket of clouds), and we had only two mosaiced scenes with such type of clouds. Given the fact that this algorithm does not use any SWIR or thermal IR bands, and is based solely on the spatial and spectral information extracted by the filters applied during training, the cloud POD rate is high, and additional training samples with such cloud types might increase the positive prediction rate of such cloud types even further. Since we are limited with IR bands, another option to improve cirrus detection with the current CNN-type algorithm is to incorporate a texture detection feature block classification (Tivive and Bouzerdoum, 2006; Cimpoi et al., 2015; Andrearczyk and Whelan, 2016) in our algorithm scheme, which can help separating the cloud types by their texture, rather than by their spectral or spatial structure. In short, the idea of a texture descriptor is to add a layer after one of the convolution layers in Block#1, where the feature maps created by the previous convolution layer are pooled by this layer by calculating the average of their activation function. This results in one single value per feature map (i.e. # of filters of the convolution layer), similar to an energy response of a filter bank of varying features. Then, the output from this layer can be concatenated with the shape features output of Block#1 and used as an additional input to Block#2.

With the combined cloud and cloud shadow algorithm, we observed a lower POD for the cloud shadows (mean of 77% with a 12% standard deviation among the mosaic scenes tested), compared to state-of-the-art (e.g. as in Foga et al., 2017 comparison). Nevertheless, the cloud-free and cloudy pixel prediction rates were not affected much by the addition of a third class, which leaves us optimistic as to the capability of the algorithm to improve upon cloud shadow predictions given more training samples. Although the mean prediction rate result of $77 \pm 12\%$ is fairly low compared to the CFmask for cloud shadow prediction (85–94% over various surfaces), we note that our cloud shadow algorithm is generated in parallel with the cloud prediction algorithm, unlike existing methods, e.g. Fmask (Zhu and Woodcock, 2012), which use the cloud BT (brightness temperature) from an IR channel, together with a assumed lapse rate and a 3D reconstruction of the shadow from the detected cloud layer. From our limited training and test sets we do see that our shadow detection is doing well for different shadow geometries, without the need to calculate them specifically from the cloud pixels. However, we do see that the cloud shadow detection is still not fine-tuned in terms that it overpredicts land areas as shadows (Fig. 13 for example).

4.2. WV-2 based CNN network implemented to predict sentinel clouds

We have trained the cloud only DeepLab CNN network using WV-2 4-channel imagery, as described in sections 3.1.1 and 3.1.2. Using this network weights, we input Sentinel image patches and predict their cloud mask using the aforementioned algorithm. In addition, we do not rescale the Sentinel images (which are 5 time lower in their spatial resolution) before their utilization in the prediction process. Although the expected performance of such an "ad-hoc" prediction was very low, we were surprised that our WV-2 based CNN scheme is doing as well, and even better in some of the cases when compared to the original Sentinel cloud mask, which was supplied with the imagery files. To test our algorithm prediction versus the original Sentinel cloud mask prediction, we have generated a limited set of 18 ground-truth images, from Sentinel imagery over the Fiji Islands, as explained in section 3.1.2. Comparing 19 512×512 patches from the various islands, we saw that the WV-2 based CNN algorithm is doing better in detecting clouds than the original cloud mask supplied with the Sentinel images (with an average true prediction rate of 81% versus 48%). Also, as seen before, our WV-2 based algorithm tends to overpredict clouds, due both to the upscaling smoothing issue and the relatively conservative training set that was used (i.e. even small, relatively thin low clouds were classified as clouds for the training purposes). In that sense, the Sentinel cloud mask has less cloud-free errors, but is often highly underpredicting clouds (see for example Fig. 14, last row). Although the augmented CNN algorithm performs fairly well, it still gets much lower cloud prediction rates compared to the model implementation on the WV-2 images. One notable issue that might cause these lower prediction rates are that WV-2 and Sentinel intensities for clouds (and other surface types) are shifted (Fig. 7), and a different normalization is needed for each of the image sets. As the network was trained on the WV-2 images, its weights and filter thresholds are optimized to the WV-2 channels intensity distributions. In our prediction process of Sentinel cloud mask using the selected CNN architecture, we experimented with several normalization procedures prior to the introduction of the Sentinel images into the network for prediction and noticed that the results are very sensitive to this step. We found that using the mean 'depth' of the image (two times the mean and standard deviation), where we subtracted the mean and divided by the input image standard deviation gives the best results. However, as this parameter had to be tuned manually, its application in an automated detection scheme is limited. Following this, we made an attempt to apply a more "automated" method that can augment the differences among the two satellite sensor domains by using the idea of DANN for training, as discussed in the next section, which theoretically should overcome these channel shift differences and will enable our algorithm to be applied on both WV-2 and Sentinel images similarly.

4.3. DANN network for cloud prediction on different platforms

The idea of domain adversarial NN (DANN) states that any two domains, which their distributions are shifted can be "aligned" in such a way that this shift will not impose an issue when trying to predict similar features from either. This shift "alignment" is being introduced into the CNN scheme through a parallel domain classifier branch, which maximize the loss between the domains so that the network eventually cannot tell the differences among them. To test this concept on our dataset and problem, we constructed a simple 2-layer CNN, with two parallel branches for classification and domain prediction, as detailed in Fig. 8. We trained the network on a limited set of images and tested it with/without the domain discriminator branch for the prediction of cloud mask from Sentinel images. We showed that although we used a simple network architecture, much inferior than the one used in our selected CNN scheme (Fig. 5), we managed to get improved prediction for both the cloudy and clear classes by using the DANN approach in comparison with the 'source' training only approach. This shows great promise in the ability to deduce common features among two different platforms, without the need to tweak their inputs separately.

4.4. Summary, conclusions and future work

In this work, we have developed a CNN based cloud and cloud shadow detection algorithm, which was originally trained on WV-2 4-channel (RGB_NIR) only imagery, for the NASA NeMO-Net project (Chirayath et al., 2018a, 2018b; Chirayath and Li, 2019). The algorithm performs well and is comparable in accuracy to existing cloud mask algorithms that were developed for Landsat. The advantage of this algorithm is that it takes into account both the spectral and the spatial context of the multi-channel imagery, and eliminates the need to set a complicated algorithm based on constant and dynamic threshold steps scheme for its application. Also, it can incorporate both cloud and cloud shadows concurrently within one framework, unlike existing algorithms. Surprisingly, our algorithm showed good results when applied to concurrent four-channel Sentinel images, with higher cloud prediction rates than the Sentinel cloud mask that was supplied with the imagery. However, to apply our prediction scheme (constructed for WV-2 data), we had to apply a nomalization procedure before input of Sentinel images for prediction, while the resulting predictions seem

sensitive to this procedure. To overcome this manual and experimental step, which might impede the application of such approach in an automatic global retrieval scheme, we tested a domain adapation approach that has a potential to eliminate this need. We constructed a simple CNN architecture to test that hypothesis and found that the domain adaptaion approach increased our capability to predict clear and cloudy pixels in Sentinel images from a network trained by WV-2 imagery alone. This result is important to our NeMO-Net project as it is planned to ingest and predict upon multi-modal satellite datasets such as Sentinel and Landsat, in addition to WV-2 and higher resolution airborne sensors. In a follow-up manuscript, we are working to implement the DANN approach presented here on both the clouds and the marine habitat classes concurrently using an extended training and test sets.

Overall, our selected CNN architecture shows great promise both in constructing a new cloud and cloud shadow detection algorithm from a limited set of channels (specifically excluding IR channels), since it capitalizes on both the spectral and spatial domians, unlike the current operational methods.

To generalize this algorithm to be applied globally there will be a need to increase the training and ground truth sets. In addition, to improve thin cirrus cloud detection capability, we suggest to apply a texture feature extractor within the CNN architecture.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2019.111446.

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